AFF–1128

Sub. Code
4MIT1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Information Technology

COMPUTER ARCHITECTURE

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Specify the algebraic function, graphic symbol and truth table of the AND gate.

2. Draw the graphic symbol and truth table of XOR gate.

3. List out the machine instructions that perform logic operations.

4. Represent the input-output configuration with the help of a diagram.

5. Mention the operations of stack.

6. State the most common fields found in instruction formats.

7. What are the differences exist between the central computer and each peripheral?

8. Enumerate the initial sequence of instruction to control the interrupt hardware.

9. Differentiate tightly coupled system from loosely coupled system.

10. Write the function table of typical RAM chip.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the control unit of basic computer with neat diagram.

Or

(b) Describe the bus and memory transfers.

12. (a) Discuss the assembly language with a program to subtract two numbers.

Or

(b) Write an assembly language program to add two double-precision numbers.

13. (a) Enumerate the characteristics of CISC.

Or

(b) List out the instruction set for Berkeley RISC I.

14. (a) Describe the addition and subtraction with signed-magnitude data.

Or

(b) With neat block diagram, Explain the DMA.

15. (a) Discuss the associative mapping of cache memory.

Or

(b) Explain the memory connection to the CPU with neat diagram.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. With graphic symbol and characteristic table, explain the operations of the different types of flip-flops.

17. Explain the micro-operations of all memory-reference instruction with neat flowchart.

18. Describe the different types of addressing modes of the instructions.

19. Discuss in detail about the division algorithms with neat flowchart.

20. Write down in detail about associative memory.

AFF–1129

Sub. Code
4MIT1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Information Technology

DATA STRUCTURES AND C++

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the advantages of an Object Oriented Programming Paradigm?

2. Define Object. How they are created?

3. What are the different forms of Inheritance?

4. Why do we need Virtual Functions?

5. List out the efficient features of an efficient Algorithm.

6. Define Stack.

7. What is meant by Binary Tree?

8. Define Graph.

9. Why Sorting is needed?

10. What is the average efficient of Heap Sort?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What is Procedure Oriented Programming? Explain its main characteristics.

Or

(b) List and explain the important properties of the Constructor.

12. (a) How single inheritance is different from multiple inheritance? Discuss.

Or

(b) Discuss briefly about Pointers with suitable examples.

13. (a) Mention and explain the qualities and capabilities of an algorithm.

Or

(b) Explain the procedure for deletion in a queue.

14. (a) How will you insert an element in a Binary Search Tree?

Or

(b) Write short notes on Minimum Spanning Tree.

15. (a) How Insertion sort is different from Shell Sort? Explain.

Or

(b) Explain the importance of Dynamic Programming.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about Operator Overloading with suitable illustrations.

17. State the importance of Exception Handling.

18. Describe the suggestions that can be useful in designing efficient algorithms.

19. Discuss in detail about the AVL Tree with neat sketch.

20. Give a detailed note on Merge Sort with appropriate example.

AFF–1130

Sub. Code
4MIT1C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Information Technology

JAVA PROGRAMMING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Define JVM.

2. What is web browser? Give an example.

3. Distinguish between private specifier and public specifier.

4. List the precedence of arithmetic operators.

5. Write any two similarities between interfaces and classes.

6. What is vector? Give the syntax.

7. What is local applet?

8. Write the uses of drawstring ( ) method in AWT?

9. Write any two features of JDBC.

10. Mention any two advantages of RMI.

Section B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain about the history of Java.

Or

(b) How is Java platform independent? Explain.

12. (a) Explain about the arithmetic and logical operators available in Java.

Or

(b) Discuss any three mathematical functions with their syntax.

13. (a) What is a constructor and how to create a constructor?

Or

(b) What is multiple inheritance? Explain with an example.

14. (a) Discuss the life cycle of Applet.

Or

(b) Explain how to create an executable applet.

15. (a) Write short notes on JSP.

Or

(b) Write the steps involved in create a Session Bean.

Section C

(3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about the various data types available in Java.

17. Discuss about operator precedence and associativity in detail.

18. Explain one dimensional and two dimensional arrays with an example.

19. Write the steps involved in the designing a web page.

20. Explain about the overview and architecture of RMI.

AFF–1131

Sub. Code
4MIT1C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Information Technology

PRINCIPLES OF E-COMMERCE

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define E-Commerce.

2. What is public policy?

3. Define Web-Security.

4. Define World Wide Web.

5. What is meant by digital token?

6. Define EDI.

7. Define macro forces.

8. What is meant by document library?

9. Define Internet.

10. What is market research?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain about the E-Commerce of media convergence.

Or

(b) Explain the Global Information Distribution Networks.

12. (a) Write short notes on Electronic Commerce.

Or

(b) Write short notes about the technology behind web.

13. (a) Explain the Digital Token and Credit Card.

Or

(b) Write short notes about EOI applications.

14. (a) Write short notes about the internal Information System.

Or

(b) Explain the corporate data warehousing.

15. (a) Write short notes on Information-Based Marketing.

Or

(b) Briefly explain the E-Commerce catalogs.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the market forces influencing I-Way.

17. Discuss about the mercantile models from the Consumer's perspectives.

18. Explain the Electronic data interchange.

19. Discuss about the Macro forces and Internal Commerce.

20. Explain the searched Resource Discovery Paradigms.

AFF–1132

Sub. Code
4MIT1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Information Technology

Elective — NUMERICAL METHODS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What is the basic principle using which Mullei's method has been established?

2. State the underlying principle of Bairstow method.

3. Explain Gauss elimination method to solve system of equation $AX = B$.

4. Explain diagonal dominance.

5. State Gregory Newton forward interpolation formula.

6. Define first, second and third order divided difference.

7. Write the formula to find y'' at $x = x_n$ from a set of values of x and y.

8. State Simpson's 1/3 rule.

9. Using Euller's method solve $y' = x + y + xy$, $y(0) = 1$ compute y at $x = 0.1$ by taking $h = 0.1$.

10. Write Milne's predictor – corrector formula.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Find the positive root of the equation $x^3 - 4x + 1 = 0$ correct to 4 places of decimals using Chebyshev's method.

Or

(b) Find the real root of $3x - \cos x - 1 = 0$ by Newton's Raphson method.

12. (a) Solve the following system of equations by Gauss elimination method :

$5x - y + z = 10$, $2x + 4y = 12$, $x + y + 5z = -1$.

Or

(b) Find the numerically largest eigen value of the matrix $A = \begin{pmatrix} 3 & -5 \\ -2 & 4 \end{pmatrix}$ by power method.

13. (a) Find the divided difference of all orders of the function $f(x) = x(x+1)(x+2)(x+3)$ corresponding the arguments 1, 2, 4, 6, 7.

Or

(b) Using Newton's forward interpolation formula find y at $x = 8$ from the table :

x :	0	5	10	15	20	25
y :	7	11	14	18	24	32

14. (a) Find the values of $f'(0)$ and $f''(0)$ from the following table :

x :	0	1	2	3	4	5
$f(x)$:	4930	5026	5122	5217	5312	5407

Or

(b) Find $\int_{-2}^{2} x^4 \, dx$ by Simpson's rule taking $h = 1$.

15. (a) Using Taylor series method solve $y' = xy + y^2$, $y(0) = 1$ at $x = 0.1$.

Or

(b) Given $y' = x^2(1+y)$, $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$, $y(1.3) = 1.979$, Evaluate $y(1.4)$ by Adam's-Bash forth method.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Find the root of the equation $\cos x = xe^x$ that lies between 0 and 1 correct to 4 places of decimals using Muller's method.

17. Solve $10x_1 + 2x_2 + x_3 = 9$, $x_1 + 10x_2 - x_3 = -22$, $-2x_1 + 3x_2 + 10x_3 = 22$ by Gauss Seidel method.

18. Fit the cubic Spline for the data

x :	1	2	3
y :	-6	-1	16

Hence evaluate $y(1.5)$ given that $y_0'' - y_2'' = 0$.

19. Evaluate the integral $I = \int_1^2 \int_1^2 \frac{dx\,dy}{x+y}$ using Trapezoidal rule with $h = k = 0.5$.

20. Using Runge-kutta method of 4th order solve $y^1 = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $x = 0.2$.

AFF–1134

Sub. Code
4MIT2C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Information Technology

COMPUTER NETWORKS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the three criteria necessary for an effective and efficient network?

2. Define Topology.

3. What is the purpose of digital modulation?

4. Why FDD is needed?

5. List out the different types of error correcting codes.

6. Define ARQ.

7. What do you mean by datagram?

8. Draw the frame format of UDP header.

9. What is mail submission?

10. Mention the purpose of content distribution network (CDN).

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Define protocol. What are the key elements of protocol?

Or

(b) Discuss the design issues for the layers.

12. (a) What is spread spectrum? Explain its types.

Or

(b) List and explain the essential elements of GSM.

13. (a) Explain the important design issues of data link layer.

Or

(b) What do you mean by CSMA/CA? Discuss its importance.

14. (a) Define routing. How routing algorithms are classified?

Or

(b) Explain the concept of shortest path algorithm with neat sketch.

15. (a) Discuss the architecture and services of E-Mail.

Or

(b) Write short notes on One-Time pad.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Compare and contrast the layers of OSI model with TCP/IP model.

17. Describe the importance of wireless transmission.

18. Discuss briefly about the bluetooth architecture and its services with neat sketch.

19. Explain in detail, how connections are established and dosed in TCP.

20. Give a detailed note on Data Encryption Standard (DES).

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AFF–1136

Sub. Code
4MIT2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Information Technology

DATA MINING AND WAREHOUSING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a data warehouse?

2. What are fact tables?

3. State the need for data cleaning.

4. Define visualization. Name two visualization tools.

5. What is a decision tree?

6. What is an Association rule?

7. Define population and mutation in the context of GA.

8. State two differences between supervised and unsupervised learning.

9. Name the four levels of temporality.

10. What is spatial mining?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain the basic OLAP operations.

Or

(b) Discuss the salient features of data modeling tools.

12. (a) Explain the measures of similarity and dissimilarity.

Or

(b) Explain the types of data.

13. (a) Explain Dynamic item set algorithm.

Or

(b) Describe classification using back propagation method.

14. (a) Describe hierarchical clustering.

Or

(b) Explain the working of GA.

15. (a) Discuss the issues in text mining.

Or

(b) Write a note on visual data mining.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the data warehouse architecture.

17. Explain the different forms of knowledge and the steps in KDD.

18. Explain Apriori algorithm.

19. Explain any two clustering algorithms.

20. Describe web content and structure mining.

AFF–1137

Sub. Code
4MIT2E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Information Technology

Elective – WIRELESS COMMUNICATIONS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List some parameters of Multipath channels.

2. What do you meant by Scattering?

3. Mention the algorithms for adaptive equalization.

4. List any two characteristics of speech signals.

5. What is channel assignment? What are the types?

6. Define cell.

7. What are the basic channels available in GSM?

8. What do you mean by forward and reverse channel?

9. Write the demerits of 3G.

10. List any four Ad hoc routing protocols.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) List the advantages and disadvantages of wireless communications.

Or

(b) Explain about Impulse response model.

12. (a) Discuss about Linear predictive coders.

Or

(b) Summarize the principles of cellular communication.

13. (a) Write a brief note on frequency reuse.

Or

(b) Explain the consideration of components of a cellular system.

14. (a) Discuss about GSM Radio subsystem.

Or

(b) Write short notes on frequency and channel specifications of CDMA.

15. (a) Explain about Wireless sensor networks.

Or

(b) Briefly discuss the concept of 4G.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain about the types of small scale fading.

17. Discuss in detail about FDMA.

18. Explain how to improve coverage and capacity in cellular systems.

19. Describe about reverse CDMA channel.

20. Write a detailed note on Mobile Ad hoc networks.

AFF–1140

Sub. Code
4MIT3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Third Semester

Information Technology

COMPILER DESIGN

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define the term Compiler.

2. What is the purpose of code generator?

3. Mention the symbol table entries.

4. What is meant by parsing?

5. What is a syntax tree?

6. Write the use of optimizing translator.

7. List the storage allocation strategies.

8. Define the term intermediate code.

9. Define run time storage management.

10. What is a flow graph?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Discuss on simple one pass compiler.

Or

(b) Write about input buffering.

12. (a) Give a note on top down parsing.

Or

(b) Discuss about SLR parsing.

13. (a) Write a note on syntax directed definitions.

Or

(b) Describe top-down translation.

14. (a) Discuss on runtime organization.

Or

(b) Write about parameter passing.

15. (a) Give a note on issues in the design of code generator.

Or

(b) Describe loops in flow graphs.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss in detail about Lexical analysis.

17. Write about shift reduce parsing.

18. Explain bottom up evaluation of S-attributed definitions and L-attributed definitions.

19. Describe about code generation.

20. Explain in detail about code optimization.

AFF–1141

Sub. Code
4MIT3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Information Technology

SOFTWARE ENGINEERING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. List out the characteristics of Software.

2. What are the various levels of Capability Maturity Model (CMM)?

3. Why Requirements Elicitation is difficult?

4. What is the purpose of PSPEC?

5. Mention the three important characteristics of goods design.

6. Why abstraction is needed?

7. Define Verification.

8. What is the role of Independent Test Group (ITG)?

9. Mention the different categories of Risk.

10. Why FTR is so important?

Part B $(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) Discuss the main objectives of Process framework.

Or

(b) What are the advantages and disadvantages of Linear Sequential Model?

12. (a) Explain the salient features of Requirement Analysis.

Or

(b) Write short notes of flow oriented Modeling.

13. (a) Illustrate the importance of functional independence.

Or

(b) How do we assess the quality of a Software Design?

14. (a) List and explain the various levels of Testing.

Or

(b) What errors are commonly found during Unit Testing? Discuss.

15. (a) Write short notes on RMMM Plan.

Or

(b) Discuss briefly about Software Reliability.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss in detail about the Evolutionary Software Process Models with neat sketch.

17. Explain briefly about Class based modeling with suitable example.

18. Give a detailed note on User Interface design.

19. Explain briefly about Black Box Testing with suitable sketch.

20. Describe the importance of Risk Management with suitable illustrations.

AFF–1142

Sub. Code
4MIT3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Information Technology

CLOUD COMPUTING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Cloud Computing with an example.

2. What are the properties of cloud computing?

3. List out the various types of communication used in cloud computing.

4. What are the various online Grocery list application?

5. How will you store a file in cloud?

6. Write some online event management application.

7. What is SaaS?

8. Write about Google Apps.

9. What is server virtualization?

10. What is storage area network?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Give an account of cloud computing today.

Or

(b) Write the advantages and disadvantage of cloud service development.

12. (a) Write short note on Collaborating on Schedules.

Or

(b) Write short note on Collaborating on Household Budgets.

13. (a) Write about various online event management applications available.

Or

(b) Write short note on file sharing.

14. (a) Write short note on Vmware vCloud.

Or

(b) Write short notes on Microsoft Office Live.

15. (a) Give the benefits of Virtualization.

Or

(b) Write short note on Hypervisor Management Software.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the working of Cloud Computing. Give it benefits.

17. Discuss in detail about the use of cloud in corporation.

18. Explain the various options available for word processing and databases.

19. Discuss in detail about Windows Azure Platform.

20. Explain in detail about Cloud Virtualization.

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AFF–1143

Sub. Code
4MIT3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Information Technology

Elective — MULTIMEDIA AND ITS APPLICATIONS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define multimedia.

2. Which information carrier will be used for information exchange between different places?

3. What is the frequency? How is it measured?

4. Mention any four video frame grabbers.

5. Specify the three categories of animation languages.

6. What are the resolution and storage capacity per image of CGA?

7. State the major steps of data compression.

8. Give the rough classification of compression techniques in multimedia.

9. List out the tools used to implement the conversational services.

10. Enumerate the design issues need to be considered when implementing editors for multimedia documents.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) What are the data stream characteristics for continuous media? Discuss.

Or

(b) What are the main properties of multimedia system? Explain.

12. (a) Explain the speech analysis.

Or

(b) Describe the speech transmission.

13. (a) Briefly discusses the important measures of visual representation.

Or

(b) Describe the computer video format.

14. (a) Discuss the H.261 video coding standard.

Or

(b) Describe image preparation step of JPEG.

15. (a) Write down about implementation of messaging and retrieval services.

Or

(b) Briefly explains the devices available for media preparation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Discuss the traditional data stream characteristics.

(b) Explain the different types of medium.

17. Discuss in detail about image synthesis.

18. Present an overview of computer-based animation.

19. Explain in detail about MPEG video coding standard.

20. Write down in detail about the different editors used for media composition.

AFF–1144

Sub. Code
4MIT3E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Information Technology

Elective – .NET TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List any two features of CLR.

2. What is a component?

3. Mention any four VB.NET looping statements.

4. What do you meant by constructor?

5. State the need of timer control.

6. Differentiate between combo boxes and check boxes.

7. What is Session object?

8. What is the basic difference between ASP and ASP.NET?

9. List any two benefits of ADO.NET.

10. What are the namespaces used in ADO.NET to connect to a database?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Summarize the benefits of .NET.

Or

(b) Make a note on .NET framework libraries.

12. (a) Explain about Inheritance in VB.NET with an example.

Or

(b) Discuss arrays in VB.NET with suitable example.

13. (a) Write a note on Text boxes.

Or

(b) Narrate about scroll bars.

14. (a) Explain about ASP.NET page directives briefly.

Or

(b) Write short notes on Collections and Lists in ASP.NET.

15. (a) Discuss about the basic concepts of ADO.NET.

Or

(b) Explain how to use SQL server with VB.NET.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Briefly explain about Visual Studio Integrated Development Environment.

17. Discuss about delegate and events in VB.NET.

18. How do you create a Windows application using Window controls? Explain.

19. Explain how to build forms with web server controls in ASP.NET.

20. Write a detailed note on data management with ADO.NET.

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AFF–1146

Sub. Code
4MIT3E4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Information Technology

Elective — COMMUNICATION SKILLS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer the following questions.

1. Define interpersonal communication.

2. What is meant by upward communication?

3. Define verbal communication.

4. How do you greet in a conversation?

5. Why do we write a memo?

6. What is a resume?

7. What is meant by numerical ability?

8. What is the goal of group discussion?

9. What is meant by critical reasoning?

10. What is the purpose of an interview?

Part B (5 × 5 = 25)

Answer the following questions.

11. (a) Explain horizontal communication.

Or

(b) Discuss the importance of communication.

12. (a) What are the soft skills?

Or

(b) Write a short note on the importance of telephone conversation.

13. (a) Describe the process of sending an E-mail.

Or

(b) What are presented in the minutes of a muting?

14. (a) How is a group discussion organised?

Or

(b) Write a short note on job interview.

15. (a) What are verbal analogies?

Or

(b) How are instructions given?

Part C (3 × 10 = 30)

Answer any **three** of the following questions.

16. What are the principles of effective communication?

17. How do you distinguish between verbal and non-verbal communication?

18. Write a job application for the position of programmer in a software company.

19. What are the skills needed for team work?

20. Write an essay on numerical estimation and numerical reasoning.